

Week 1 & 2 Insight Summary

Titanic Dataset

The exploratory data analysis revealed several factors that influenced passenger survival on the Titanic. Gender was one of the most important factors, with female passengers having a much higher survival rate than male passengers. Passenger class also played a significant role, as first-class passengers were more likely to survive compared to those travelling in second and third class.

The analysis also showed that passengers who paid higher fares generally had better survival outcomes. While age appeared to have some influence, the difference between survivors and non-survivors was not as pronounced as that of gender and passenger class. A few outliers were observed in variables such as Age and Fare, but they were retained since they represent valid observations rather than data quality issues.

Overall, the data cleaning process successfully addressed missing values and confirmed that there were no duplicate records affecting the analysis.

House Prices Dataset

The housing dataset showed that the size of a house is one of the strongest factors influencing its price. As the floor area increased, property prices generally increased as well. Houses with more bathrooms also tended to have higher prices, while the relationship between the number of bedrooms and price was positive but less pronounced.

Properties with additional features such as air conditioning, guest rooms, preferred locations, and better furnishing status were generally more expensive than those without these features. Some high-priced properties appeared as outliers, but they were retained because they represent premium homes and provide useful information for analysis.

The correlation analysis indicated that variables such as area, bathrooms, stories, and parking have meaningful relationships with house prices and are likely to be important features in predictive modelling.

Conclusion

This assignment provided practical experience in data cleaning and exploratory data analysis using real-world datasets. The analysis demonstrated how proper data preparation helps uncover meaningful patterns and relationships before model development. The insights gained from both datasets highlight the importance of understanding the data before applying machine learning techniques. Overall, the cleaned datasets are well prepared for future predictive modelling tasks.